1/10/2026

LAB NO 04

Operating System Lab

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Submitted By: Wasif Ali

Reg ID: 245187

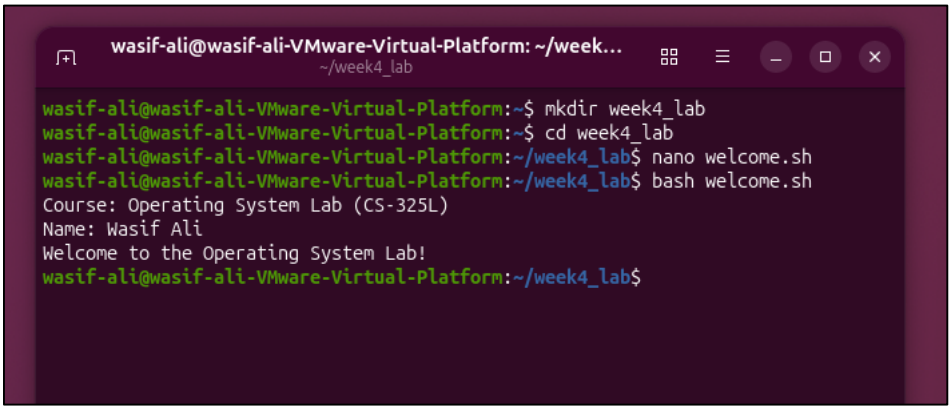
Class: BSCS F'24 B

LAB TASK

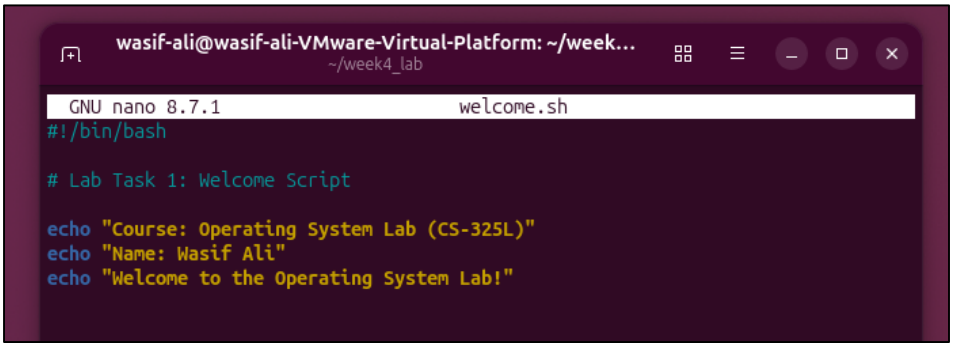
Lab Task 1 - Welcome Script

1. **Use** Bash to create a script named welcome.sh.
2. **Complete** the script so it displays the course name, your name, and a welcome message.
3. **Show** the script output in the terminal.

Commands:



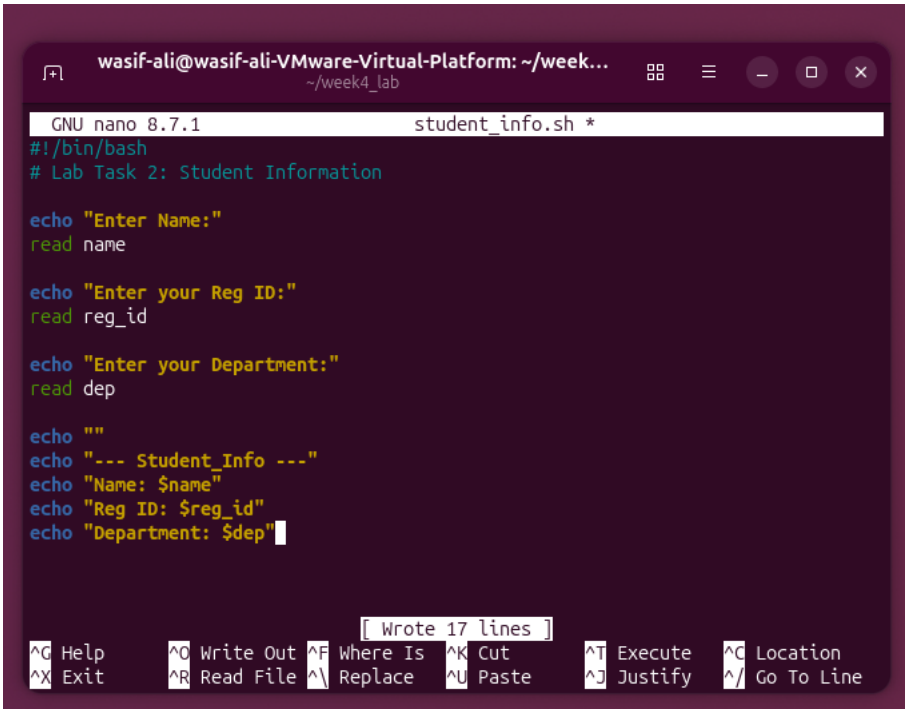
Script File:



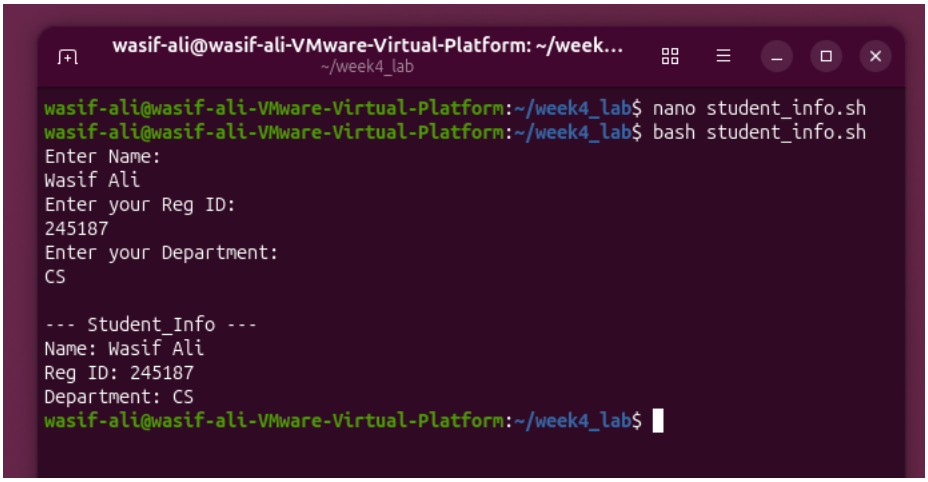
Lab Task 2 - Student Information

1. **Produce** a script named student_info.sh that asks for the student name, registration number, and department.
2. **Use** read statements to store each input in a variable.
3. **Demonstrate** the script with sample information.

Script File:



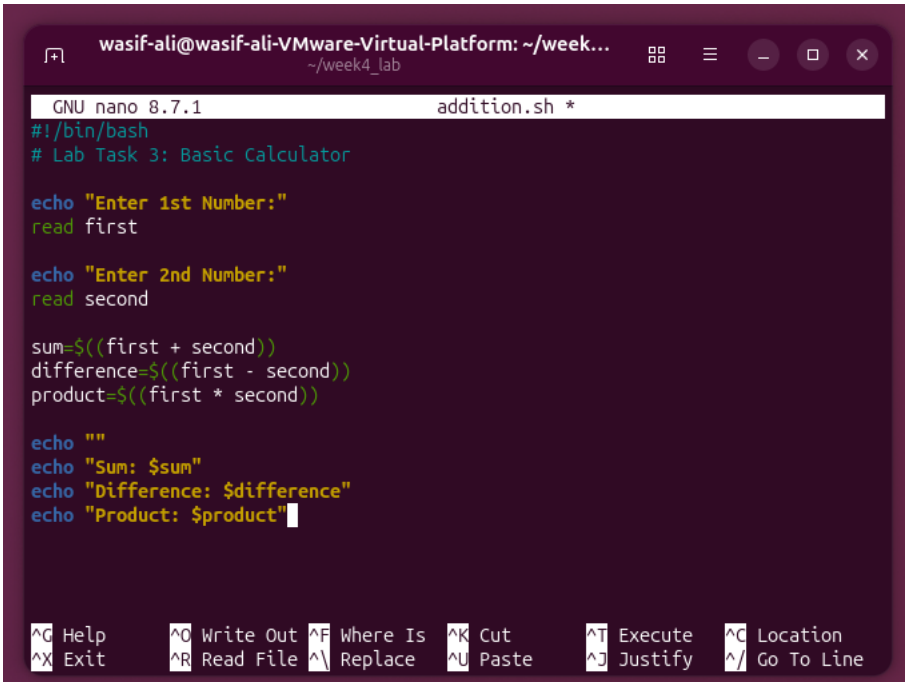
Commands:



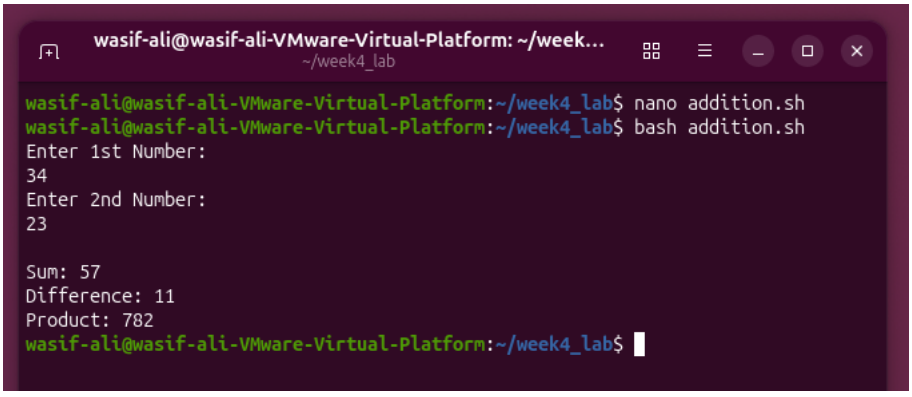
Lab Task 3 - Basic Calculator

1. **Solve** the addition task by creating a script named addition.sh that reads two integers and displays their sum.
2. **Change** the script so it displays the difference and product as well.
3. **Report** the input values and output in your lab record.

Script File:



Commands:



Lab Task 4 - Script Permissions

1. **Complete** the script execution steps with both bash filename.sh and ./filename.sh.
2. **Use** chmod +x when executable permission is required.
3. **Relate** the permission message to Linux file access.

Direct Execution Attempt (Without execute permission):

```
wasif-ali@wasif-ali-VMware-Virtual-Platform:~/week4_lab$ ./welcome.sh
bash: ./welcome.sh: Permission denied
```

When executing with ./filename.sh, the Linux kernel checks the file's execute permission bit (x). By default, newly created text files only receive read and write permissions (rw-).

Running via Interpreter:

```
wasif-ali@wasif-ali-VMware-Virtual-Platform:~/week4_lab$ bash welcome.sh
Course: Operating System Lab (CS-325L)
Name: Wasif Ali
Welcome to the Operating System Lab!
```

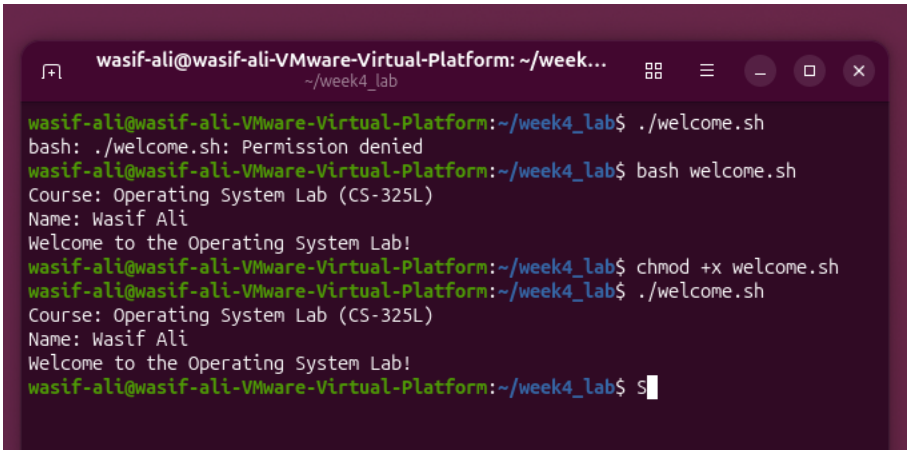
The script executes without an error because bash only needs read permission to open and interpret the file contents.

Granting Execute Permission:

```
wasif-ali@wasif-ali-VMware-Virtual-Platform:~/week4_lab$ chmod +x welcome.sh
wasif-ali@wasif-ali-VMware-Virtual-Platform:~/week4_lab$ ./welcome.sh
Course: Operating System Lab (CS-325L)
Name: Wasif Ali
Welcome to the Operating System Lab!
wasif-ali@wasif-ali-VMware-Virtual-Platform:~/week4_lab$
```

The script runs directly via ./welcome.sh because the chmod +x command adds executable permissions to the file mode flags.

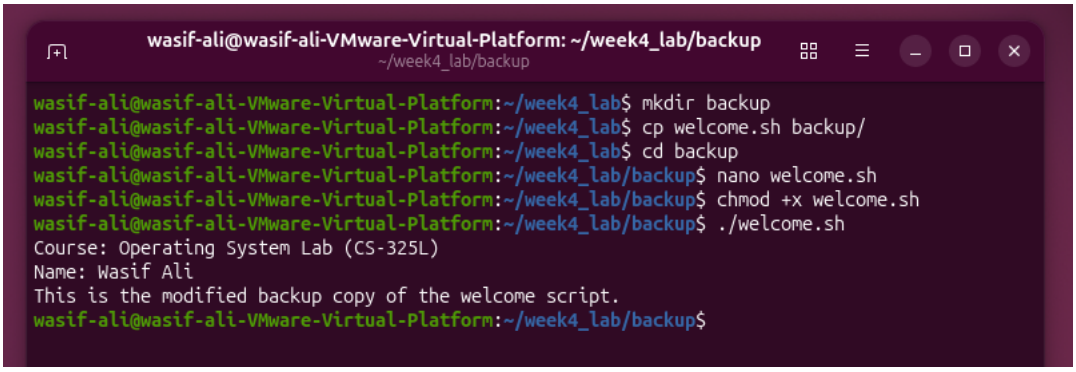
Commands:



Lab Task 5 - Organizing Script Files

1. **Prepare** a directory named week4_lab for your scripts and output notes.
2. **Transfer** a copy of welcome.sh into a subdirectory named backup with the cp command.
3. **Manipulate** the welcome message in the copied script and run it to confirm the change.
4. **Administer** executable-file permissions with chmod +x, then verify that the script can run.

Commands:



Script File:

